
title: Library Management System Project Report author: Karan Tiwari date: August 2026

1. Project Title

Library Management System

2. Problem Statement

Managing a library's catalog, members, and book loans manually is error-prone, time-consuming, and inefficient. Librarians struggle to track overdue books, generate accurate reports, and manage member subscriptions. The lack of an automated system leads to lost books, delayed notifications, and poor user experience for library members.

3. Objectives

- To automate the daily operations of a library including catalog management and circulation.
- To provide a secure, role-based system for Administrators, Librarians, and Members.
- To implement automated tracking of book loans and overdue notifications.
- To generate exportable reports for data analysis.
- To ensure data integrity, maintain an activity audit log, and provide a user-friendly interface.

4. System Architecture

The application follows a standard modern multi-tier architecture using Spring Boot.

```
graph TD
    Client[Web Browser Client] -->|REST API JSON| Controller[REST Controllers]
    Controller -->|DTOs| Service[Service Layer]
    Service -->|Entities| Repository[Spring Data JPA Repositories]
    Repository -->|SQL| Database[(Relational Database)]
```

- **Presentation Layer:** HTML/TailwindCSS frontend interacting via REST APIs.
- **Controller Layer:** Spring MVC REST Controllers handling HTTP requests and input validation.
- **Service Layer:** Contains core business logic and transaction management.
- **Data Access Layer:** Spring Data JPA repositories interfacing with the database.

5. Database Design (ER Diagram)

The relational database schema is designed to ensure normalization and data integrity.

```
erDiagram
    USER {
        Long id PK
        String username
        String password
        String email
        String fullName
    }
    ROLE {
        Long id PK
        String name
    }
    BOOK {
        Long id PK
        String title
        String isbn
        int publicationYear
    }
    AUTHOR {
        Long id PK
        String name
        String bio
    }
    CATEGORY {
        Long id PK
        String name
        String description
    }
    MEMBER {
        Long id PK
        String email
```

```

        String fullName
        String phone
    }
    LOAN {
        Long id PK
        LocalDate loanDate
        LocalDate dueDate
        String status
    }
    AUDIT_LOG {
        Long id PK
        String action
        String username
        LocalDateTime timestamp
    }

    USER }|--|{ ROLE : "has"
    BOOK }|--|{ AUTHOR : "written by"
    BOOK }|--|{ CATEGORY : "belongs to"
    LOAN }|--|{ BOOK : "borrows"
    LOAN }|--|{ MEMBER : "borrowed by"

```

6. Application Features

- **Complete CRUD Operations:** Comprehensive management of Books, Authors, Categories, and Members.
- **Authentication & Authorization:** Secure JWT-based login with Role-Based Access Control (Admin, Librarian, Member).
- **Loan Management & Scheduling:** Tracks borrowed books and automatically marks them overdue using Spring Scheduled Tasks.
- **Email Notifications:** Automatically emails members when their books become overdue.
- **Reporting:** Export functionality for Loan data (PDF) and Book data (CSV).
- **Audit Logging:** Tracks system actions (e.g., adding books, changing loan status) utilizing Spring AOP.
- **Advanced Search & Filtering:** Uses Spring Data Specifications and Pagination for efficient catalog browsing.

7. Screenshots of the Application

Login Screen

The secure authentication portal for accessing the system.  Login Screen

Admin Dashboard

Provides a high-level overview of system metrics, total books, and recent activity.

 Admin Dashboard

Books Catalog

Allows users to browse, search, and manage the library's book inventory.

 Books Catalog

8. Technologies Used

- **Backend Framework:** Java 17, Spring Boot 3.x
- **Data Access & Database:** Spring Data JPA, Hibernate, MySQL / H2 Database
- **Security:** Spring Security, JSON Web Tokens (JWT)
- **Frontend:** HTML5, Vanilla JavaScript, TailwindCSS
- **Reporting & Utilities:** OpenPDF (PDF Export), OpenCSV (CSV Export), Spring Mail (Email Notifications)
- **API Documentation:** SpringDoc OpenAPI (Swagger UI)
- **Build Tool:** Apache Maven

9. Conclusion

The Library Management System successfully fulfills all objectives by providing a robust, scalable, and user-friendly application. By leveraging the advanced features of the Spring Boot ecosystem—such as Spring Data JPA for seamless database integration, Spring Security for robust protection, and Spring AOP for audit logging—the system effectively

automates complex library workflows and provides an excellent foundation for future enhancements.